

Summary

This course offers an introduction to network analysis and is designed to provide students with an overview of the core data scientific skills required to analyze complex networks. Through hands-on lectures, labs, and projects, students will learn actionable skills about network analysis techniques using Python (in particular, the `networkx` library). The course includes network data collection, data input/output, network statistics, dynamics, and visualization. Students also learn about random graph models and algorithms for computing network properties like path lengths, clustering, degree distributions, and community structure. In addition, students will develop web scraping skills and will be introduced to the vast landscape of software tools for analyzing complex networks. The course ends with a large-scale final project that demonstrates the proficiency of the students in network analysis.

Course Learning Outcomes

- Write and maintain software for analyzing networks from a range of data sources.
- Implement network algorithms for measuring clustering, path lengths, spectra, and dynamics.
- Code generative network models for network creation and inference from data.
- Collect network data online through web scraping, incorporating data collection ethics and norms.

Course Materials

Our main resource for the course is the ever-growing Jupyter Book, where we will host all the Python notebooks used in this class <https://github.com/milo-trujillo/CNET5051>. Other than that, there are no required materials for this course, but we will periodically draw from:

- Bagrow & Ahn (2024). *Working with Network Data: A Data Science Perspective*. Cambridge University Press; 1st Edition; 978-1009212595. <https://www.cambridge.org/network-data>.

Additionally, we recommend engagement with other useful network science and/or Python materials:

- Barabási (2016). *Network Science*. Cambridge University Press; 1st Edition; 978-1107076266. <http://networksciencebook.com/>.
- Newman (2018). *Networks: An Introduction*. Oxford University Press; 2nd Edition; 978-0198805090. <https://global.oup.com/academic/product/networks-9780198805090>.
- Barrat, Barthélemy & Vespignani (2008). *Dynamical Processes on Complex Networks*. Cambridge University Press; 1st Edition; 978-0511791383. <https://doi.org/10.1017/CB09780511791383>.
- VanderPlas (2019). *Python Data Science Handbook*. O'Reilly Media, Inc; 978-1491912058. <https://github.com/jakevdp/PythonDataScienceHandbook>.

Instructor

Milo Trujillo is a Postdoctoral Research Fellow at the Vermont Complex Systems Institute and an adjunct lecturer at the Network Science Institute. His primary research interest is in studying how groups organize, govern, and interact, through patterns in our social artifacts. This work has included content moderation and deplatforming, law enforcement behavior, the emergence of alt-tech, decentralized social platforms, and the governance of open source software. Dr. Trujillo received a PhD in Complex Systems and Data Science in 2024 from the University of Vermont, and received M.S. and B.S. degrees in computer science and a B.S. in Science and Technology Studies from Rensselaer Polytechnic Institute in 2020 and 2018. Website: backdrifting.net.

Coursework, Class Structure, Grading

This is a twice-weekly hands-on class that emphasizes building experience with coding. This does not necessarily mean every second of every class will be live-coding, but it will inevitably come up in how the class is taught. We are often on the lookout for improving the pedagogical approach to this material, and we would welcome feedback on class structure. We will try to incorporate a 5-minute break for stretching, Q&A, grabbing water, etc. within each class. The course will be primarily taught by Dr. Milo Trujillo, as well as outside experts. Grading in this course will be as follows:

- Class Attendance & Participation: 10%
- Problem Sets: 45%
- Mid-Semester Project Presentation: 15%
- Final Project — Presentation & Report: 30%

Final Project

The final project for this course will demonstrate students' abilities to gather data, analyze networks, and visually present results. Students will show competency in selecting and utilizing appropriate network statistics to describe salient features of their chosen networks.

Schedule

A preliminary course schedule is on the following page. Topics may be removed to allow greater focus on core material.

	Date	Class	Instructor(s)
Week 1	Mon, Sep 7, 26	Labor Day (no class)	
	Thu, Sep 12, 26	Class 0: Introduction to the Course & Computing Setup	Milo Trujillo
	Fri, Sep 13, 26		
Week 2	Mon, Sep 14, 26	Class 1: Python Refresher (Data Structures, Numpy)	Milo Trujillo
	Thu, Sep 17, 26	Class 2: Networkx 1 — Loading Data, Basic Statistics	Milo Trujillo
	Fri, Sep 18, 26		
Week 3	Mon, Sep 21, 26	Class 3: Networkx 2 — Graph Algorithms	Milo Trujillo
	Thu, Sep 24, 26	Class 4: Distributions of Network Properties & Centralities	Milo Trujillo
	Fri, Sep 25, 26	<i>First assignment announced</i>	
Week 4	Mon, Sep 28, 26	Class 5: Scraping Web Data 1 - BeautifulSoup, HTML, Pandas	Milo Trujillo
	Thu, Oct 1, 26	Class 6: Scraping Web Data 2 - Networks from Scraped Data	Milo Trujillo
	Fri, Oct 2, 26		
Week 5	Mon, Oct 5, 26	Class 7: Clustering & Community Detection 1 - Traditional	Milo Trujillo
	Thu, Oct 8, 26	Class 8: Data Science 1 - Pandas, SQL, Regressions	Milo Trujillo
	Fri, Oct 9, 26	<i>First assignment due Oct 9</i>	
Week 6	Mon, Oct 12, 26	Class 9: Machine Learning 1 - General	Milo Trujillo
	Thu, Oct 15, 26	Class 10: Network Sampling Theory	Milo Trujillo
	Fri, Oct 16, 26	<i>Second assignment announced</i>	
Week 7	Mon, Oct 19, 26	Class 11: Visualization 1 - Python + Gephi	Milo Trujillo
	Thu, Oct 22, 26	Class 12: Sampling Networks	Milo Trujillo
	Fri, Oct 23, 26		
Week 8	Mon, Oct 26, 26	Class 13: Project Update Presentations	Everyone
	Thu, Oct 29, 26	Class 14: Dynamics on Networks 1 — Diffusion and Random Walks	Milo Trujillo
	Fri, Oct 30, 26	<i>Second assignment due October 30</i>	
Week 9	Mon, Nov 2, 26	Class 15: Dynamics on Networks 1 — Diffusion and Random Walks	Milo Trujillo
	Thu, Nov 5, 26	Class 16: Visualization 2	Prof. Pedro Cruz
	Fri, Nov 6, 26	<i>Third assignment announced</i>	
Week 10	Mon, Nov 9, 26	Class 17: Dynamics on Networks 2 — Compartmental Models	Milo Trujillo
	Thu, Nov 12, 26	Class 18: Dynamics on Networks 3 — Agent-Based Models	Milo Trujillo
	Fri, Nov 13, 26		
Week 11	Mon, Nov 16, 26	Class 19: Dynamic on Networks 4 - Temporal Networks	Milo Trujillo
	Thu, Nov 19, 26	Class 20: Statistics for Social Networks: ERGMs	Milo Trujillo
	Fri, Nov 20, 26	<i>Third assignment due Nov 20</i>	
Week 12	Mon, Nov 23, 26 Thu, Nov 26, 26 Fri, Nov 29, 26	Thanksgiving Break (no class)	
Week 13	Mon, Nov 30, 26	Class 21: Instructor's Choice: Spatial Data, OSMNX, GeoPandas	Milo Trujillo
	Thu, Dec 3, 26	Class 22: Instructor's Choice: Hypergraphs, Big Data	Milo Trujillo
	Fri, Dec 4, 26		
Week 14	Mon, Dec 7, 26	Class 23: Final Presentations	Everyone
	Thu, Dec 10, 26	Class 24: Wiggle Room Class	Milo Trujillo
	Fri, Dec 11, 26		
Week 15	Mon, Dec 14, 26	Class 25: Wiggle Room Class	Milo Trujillo
	Thu, Dec 17, 26		
	Fri, Dec 18, 26		